A Minimal Copper-Plate Storage-Scheduling Benchmark for Differential Dynamic Programming

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I. Copper-Plate Storage-Scheduling Model

A. Optimization Problem

A single storage device and a single grid connection serve a known demand p_L^t over T periods of length Δt, all quantities in per unit. Following the sign convention of [1], $P_B^t > 0$ discharges and $P_B^t < 0$ charges; B^t is the energy at the *start* of period t, so B^1 is given and B^{T+1} is terminal. With $t \in \mathcal{T} = \{1, \dots, T\}$:

$$\begin{aligned}
\min_{P_{\text{Subs}},\, P_B,\, B} \quad & \sum_{t=1}^{T} \Big[c^t P_{\text{Subs}}^t + C_B \big(P_B^t\big)^2 \Big] \Delta t \\
& \quad + \; w\big(B^{T+1} - B^1\big)^2 && \text{(1a)} \\
\text{s.t.} \quad & P_{\text{Subs}}^t + P_B^t - p_L^t = 0, \qquad t \in \mathcal{T} && \text{(1b)} \\
& B^1 = B_0 && \text{(1c)} \\
& B^{t+1} = B^t - P_B^t \Delta t, \qquad t \in \mathcal{T} && \text{(1d)} \\
& P_{\text{Subs}}^t \geq 0, \qquad t \in \mathcal{T} && \text{(1e)} \\
& P_B^{\min} \leq P_B^t \leq P_B^{\max}, \qquad t \in \mathcal{T} && \text{(1f)} \\
& B^{\min} \leq B^{t+1} \leq B^{\max}, \qquad t \in \mathcal{T} && \text{(1g)}
\end{aligned}$$

The *base instance* is (1a)–(1d), with all three bounds dropped.

B. Optimal-Control Form

Restated over $N = T$ stages in the form required by DDP:

$$\begin{aligned}
\min_{x,u} \quad & \sum_{t=1}^{N-1} \ell\big(x^t, u^t\big) + \ell^F\big(x^N, u^N\big) \\
\text{s.t.} \quad & x^1 = \hat{x}^1, \\
& x^{t+1} = f\big(x^t, u^t\big), \quad t \in \{1, \dots, N-1\}, \\
& c\big(x^t, u^t\big) = 0, \quad t \in \{1, \dots, N\}.
\end{aligned} \tag{2}$$

with state, control and initial condition

$$x^t = \big(B^t,\ \tau^t\big)^\top \in \mathbb{R}^2, \qquad u^t = \big(P_{\text{Subs}}^t,\ P_B^t\big)^\top \in \mathbb{R}^2, \tag{3}$$

$\hat{x}^1 = (B_0,\, 1)^\top$. Period-varying data is carried by the time index τ through the Lagrange interpolants on the nodes $\tau = 1, \dots, T$,

$$p_L(\tau) = \sum_{k=1}^{T} p_L^k \prod_{\substack{m=1 \\ m \neq k}}^{T} \frac{\tau - m}{k - m}, \tag{4}$$

and likewise for the price $c(\tau)$. The stage data is then

$$\begin{aligned}
f(x, u) &= \big(x_1 - u_2 \Delta t,\ \ x_2 + 1\big)^\top, && (5) \\
c(x, u) &= u_1 + u_2 - p_L(x_2), && (6) \\
\ell(x, u) &= \big[c(x_2)\, u_1 + C_B\, u_2^2 \big] \Delta t, && (7) \\
\ell^F(x, u) &= \big[c^T u_1 + C_B\, u_2^2 \big] \Delta t \\
&\quad + w\big(x_1 - u_2 \Delta t - B_0\big)^2. && (8)
\end{aligned}$$

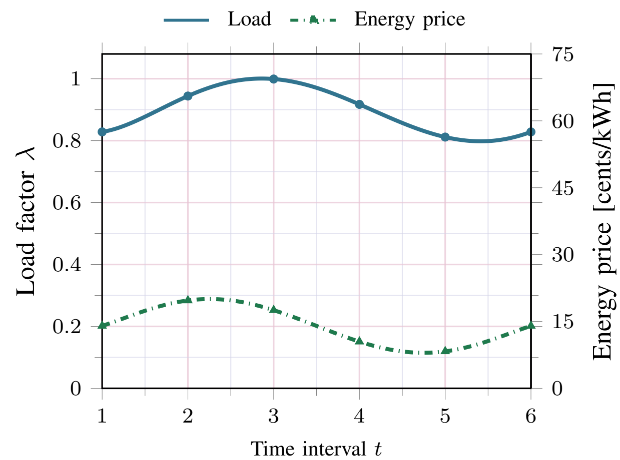

Fig. 1. Input profiles over the $T = 6$ horizon: load factor on the left axis, energy price on the right, each specified at the T horizon points. Markers are the data; the curves are the Lagrange interpolants (4) through them, and it is the curves, not the markers, that the solver evaluates. The two agree exactly at every node, by construction, but between nodes they need not be well behaved and already are not — the load interpolant reaches 0.998 against a node maximum of 0.9, and the price interpolant 69.7 cents/kWh against 62. Nothing samples those excursions here, because the time-index state advances by exactly one per stage and so only ever lands on a node; but the interpolant's *derivatives* at the nodes are what the backward pass consumes, and its degree is $T - 1$. This is the device that does not survive a realistic horizon.

Bounds (1e)–(1f) map onto control limits $b^L \leq u^t \leq b^U$. The state bound (1g) is appended as a second row of c via a slack s^t adjoined to the control, giving $n_c = 2$, $n_u = 3$:

$$\big(x_1 - u_2 \Delta t\big) - B^{\min} - s^t = 0, \qquad 0 \leq s^t \leq B^{\max} - B^{\min}. \tag{9}$$

C. Closed-Form Reference

The base instance is a strictly convex equality-constrained QP. Eliminating $P_{\text{Subs}}^t = p_L^t - P_B^t$ and $B^{T+1} - B^1 = -\Delta t \sum_t P_B^t$ reduces it to

$$\begin{aligned}
J(P_B) = \text{const} - \Delta t \sum_t c^t P_B^t + C_B \Delta t \sum_t \big(P_B^t\big)^2 \\
+ w(\Delta t)^2 s^2,
\end{aligned} \tag{10}$$

with $s = \sum_t P_B^t$. Stationarity divided through by Δt is $-c^k + 2C_B P_B^k + 2w\Delta t\, s = 0$; summing over k collapses it to

$$s = \frac{\sum_t c^t}{2C_B + 2Tw\Delta t}, \qquad P_B^k = \frac{c^k - 2w\Delta t\, s}{2C_B}. \tag{11}$$

References

[1] A. R. Jha, S. Paul, and A. Dubey, "Analyzing the performance of linear and nonlinear multi-period optimal power flow models for active distribution networks," in *2025 IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCE)*. IEEE, 2025, pp. 04–06.

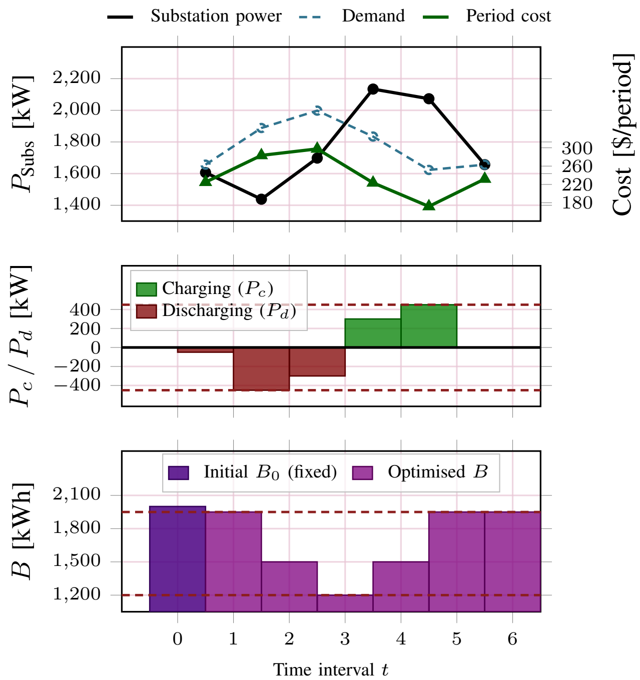

Fig. 2. The benchmark solved at $T = 6$ with all bounds (1e)–(1g) imposed. Battery actions use the convention of the tADMM study: charging is drawn upward in green and discharging downward in maroon, so a bar's direction reads as the device's action rather than as the sign of P_B^t. Read it as one worked example. The battery discharges through $t = 1$–3, hitting its 450 kW rail at $t = 2$ where the price peaks, and recharges through the two cheap intervals $t = 4, 5$, hitting the -450 kW rail at $t = 5$ where energy is cheapest. That is why the substation trace in the top panel crosses the demand it serves: it is pulled down to 1438 kW at the demand-and-price peak and pushed up to 2073 kW in the cheap trough, where it is buying energy for the device rather than for the load. Substation import is unlimited above, so the schedule is bounded only by the device, and it rides both of its limits: stored energy is drawn right down onto $B^{\min} = 1200$ kWh after the interval-3 discharge and pushed back onto $B^{\max} = 1950$ kWh at the end (dashed rules in the lower two panels). Those bounds, together with the two power rails, are exactly the active set the independent reference solver reports. Fig. 3 shows the same solution as a balance of supply against demand.

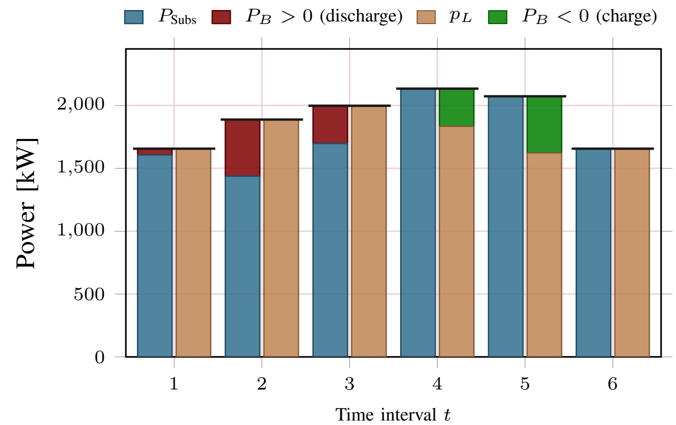

Fig. 3. Where every kW comes from and goes, at $T = 6$ with all bounds imposed. Each interval carries two stacked bars: on the left what *supplies* the node, substation import P_{Subs}^t plus battery discharge; on the right what *consumes* it, the served demand p_L^t plus battery charging. The two must reach the same height — that equality *is* the power balance (1b), and the black rule marks the level they share. Read the figure as one worked example. In $t = 2$ the price peaks at 0.197 $/kWh, and the battery discharges at its 450 kW rail, so the maroon block supplies 23.8% of the 1888 kW the node needs and the substation is held down to 1438 kW even though demand is near its own peak. Three intervals later the sign flips: at $t = 5$ the price bottoms at 0.083 $/kWh, the battery charges at -450 kW, and that charging appears on the *demand* side, pushing substation import up to 2073 kW to serve only 1623 kW of actual load. The device is doing arbitrage in the open. Over the horizon the substation imports 10 606 kWh against 10 657 kWh of load served, the 51 kWh difference being the net the battery gives back as it returns from $B = 1200$ kWh to 1950 kWh. Bars are in kW on the tADMM base, 1 p.u. = 1000 kW.

TABLE I
INSTANCE DATA

Interval t	1	2	3	4	5	6
Horizon $T = 6$						
Demand p_L^t	1.657	1.888	1.998	1.834	1.623	1.657
Energy price c^t	0.140	0.197	0.175	0.105	0.083	0.140
Horizon $T = 3$ — resampled, *not* a prefix of the above						
Demand p_L^t	1.657	1.943	1.657		—	
Energy price c^t	0.140	0.140	0.140		—	

Period length Δt	1	Initial energy B_0	2.0
Throughput C_B	0.05	Terminal weight w	5.0

Bounds, constrained variants only	$T = 3$	$T = 6$
$P_B^{\min}$, $P_B^{\max}$	-0.5, 0.004	-0.45, 0.45
$B^{\min}$, $B^{\max}$	1.9895, 1.9965	1.20, 1.95

All quantities are in per unit on the tADMM base, 1 p.u. = 1000 kW and 1 p.u.h = 1000 kWh; with $\Delta t = 1$, power and energy are numerically interchangeable. Figs. 2 and 3 report battery quantities in kW and kWh rather than per unit. At $T = 6$ the bounds below are ± 450 kW and 1200–1950 kWh. The series are the tADMM load and price profiles, $p_L^t = 2\,\lambda^t$ with $\lambda^t = 0.8 + 0.1(\sin(\theta^t - 0.8) + 1)$ and $c^t = 0.08 + 0.06(\sin\theta^t + 1)$, sampled at $\theta^t \in \text{range}(0, 2\pi, T)$. They therefore *resample* with T: the $T = 3$ row is not the first three entries of the $T = 6$ row. Two consequences of that sampling are worth stating plainly. First, both endpoints of $[0, 2\pi]$ are included, so $c^1 = c^T$ for every T. Second, at $T = 3$ the samples fall at 0, π and 2π, where sin vanishes, so *the price is constant* and the $T = 3$ instance carries no arbitrage signal at all — its optimal P_B^t is the same in every period and is set purely by the terminal penalty. The -0.8 phase offset on the load is what keeps the two series from being collinear; their Pearson correlation is 0.644 at $T = 6$. Bounds apply only to the constrained variants. Figs. 1, 2 and 3 all show the $T = 6$ case.